

Hemodynamics and Circulation

Principles and Practices of Osteopathic Medicine II



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Learning Objectives

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Learning Objective 1. Describe hemodynamics. Explain the relationship between **blood flow**, **velocity**, **pressure**, **vascular resistance** and **cross-sectional area** of blood vessel.

Learning Objective 2. Describe blood pressure and its components, including cardiac output and total peripheral resistance

Learning Objective 3. Predict factors that can alter cardiac output, stroke volume and pulse pressure.

Learning Objective 4. Evaluate and determine cardiac output given laboratory and physical findings.

Learning Objective 5. Describe Poiseuille's law and its relationship with **blood flow** and describe the La Place Law and its relevance to the cardiovascular system.

Hemodynamics

Hemodynamics refers to the principles that govern blood flow in the cardiovascular system. It refers to the principles that drives blood flow in the cardiovascular system. Velocity, Pressure, Cross-Sectional Area, Resistance and **capacitance or compliance** (capacity to hold blood).

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Fluid Movement

Fluid Dynamics

What is fluid?

Examples:

Water



oil



Blood



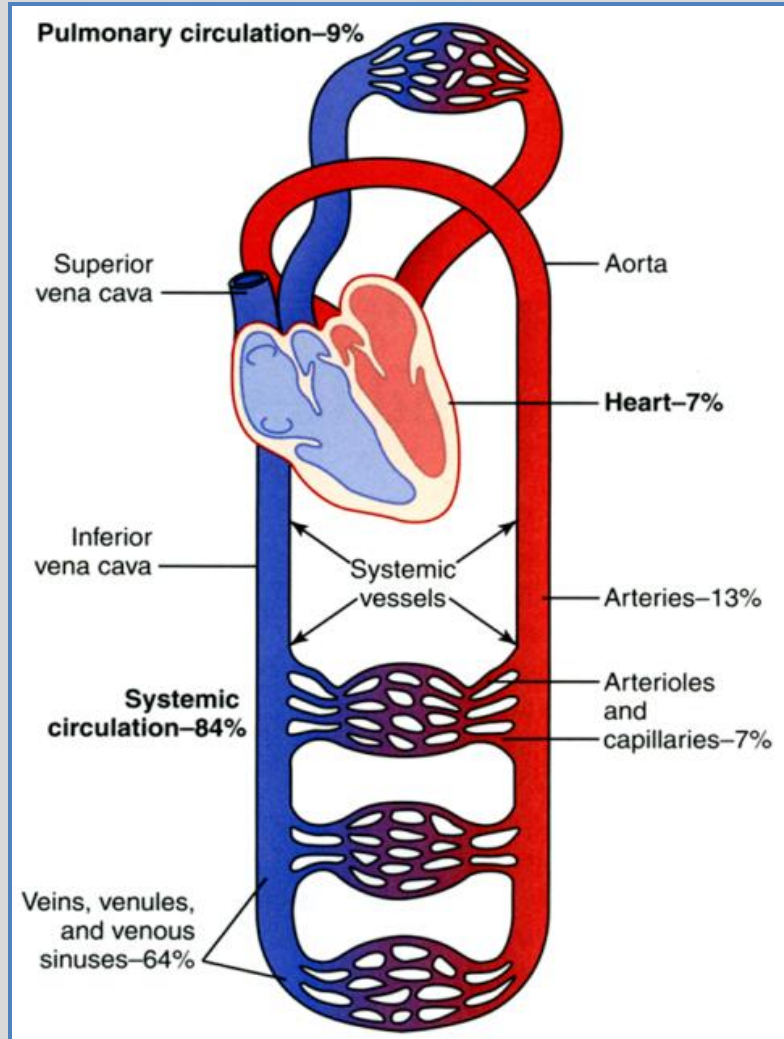
Honey



Physical Characteristics of Circulation

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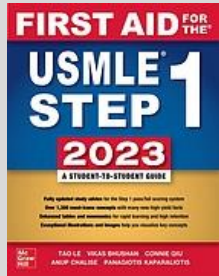
- 1) Set of closed vessels. **“Network of pipes”**
- 2) Pressure Pump (heart).
- 3) Circulating fluid (blood).

♥ **Stroke Volume (SV)** is the volume of blood pumped out of the heart with each beat

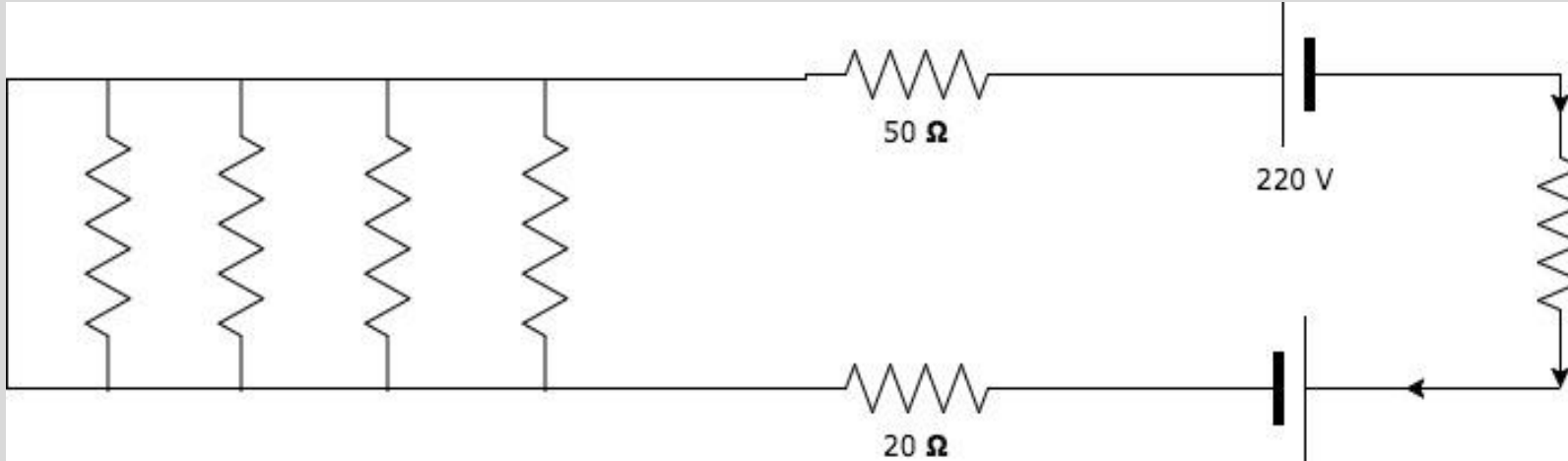
♥ **Cardiac Output (CO)** is volume of blood pumped by the heart per minute.

$CO = \text{stroke volume (mL)} \times \text{heart rate (beats/min)}$

♥ Cardiac Output is ~5 L/min.



The Vasculature = Network of Pipes



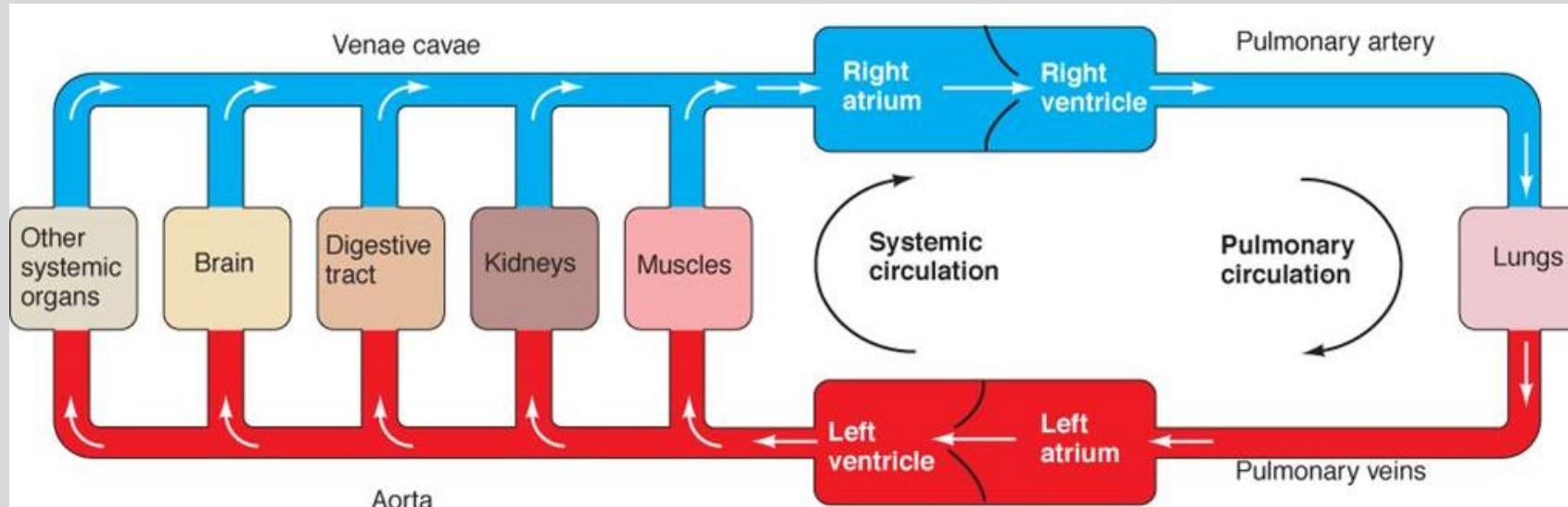
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♥ Blood is the current

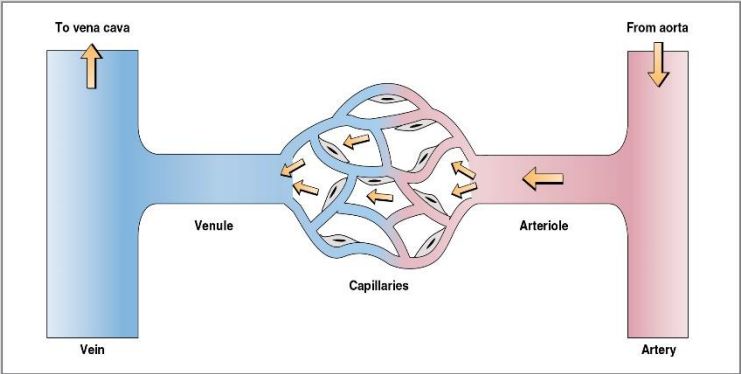
♥ Heart is the battery

♥ What are the wires?



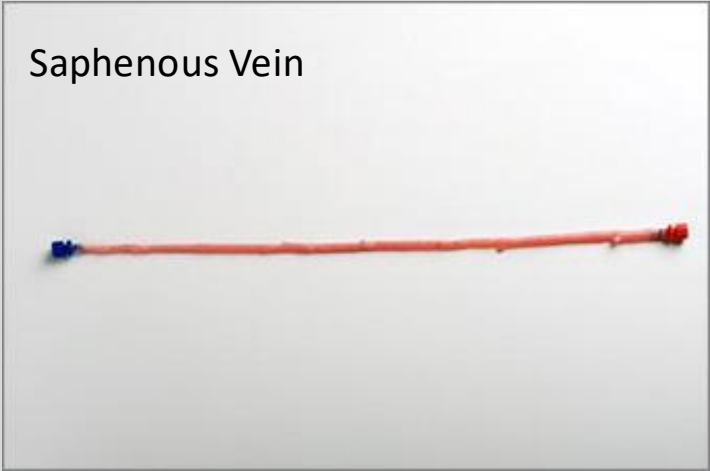
Therefore, cardiac output from the heart equals venous return to the heart.

The Vasculature = Network of Pipes



	Diameter	Medial layer thickness	Endothelium	Elastic Tissue	Muscle Tissue	Fibrotic Tissue
Arteries	4,0 mm	1,0 mm				
Arterioles	30,0 μm	6,0 μm				
Capillary	8,0 μm	0,5 μm				
Small Veins	20,0 μm	1,0 μm				
Veins	5,0 mm	0,5 mm				

1 μm (micrometer) = 0.001 mm



Basic Terminology in Circulation

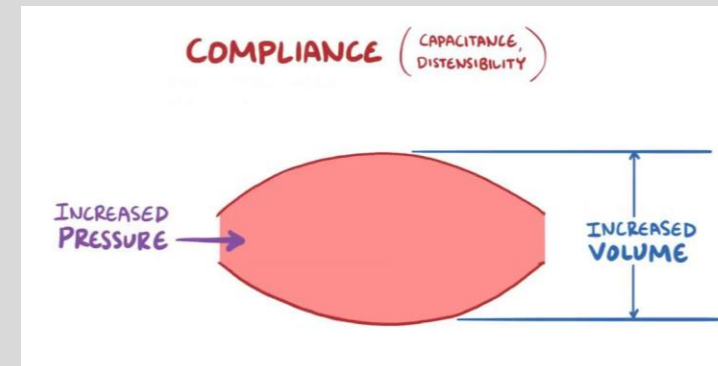
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Blood Flow is the volume of blood that passes a given point in the circulation in a given period of time (mL/sec or cm^3/sec). **volume/time**

Velocity measures the distance moved in a given amount of time. It is the distance of displacement of blood per unit time (cm/sec). **distance/time**

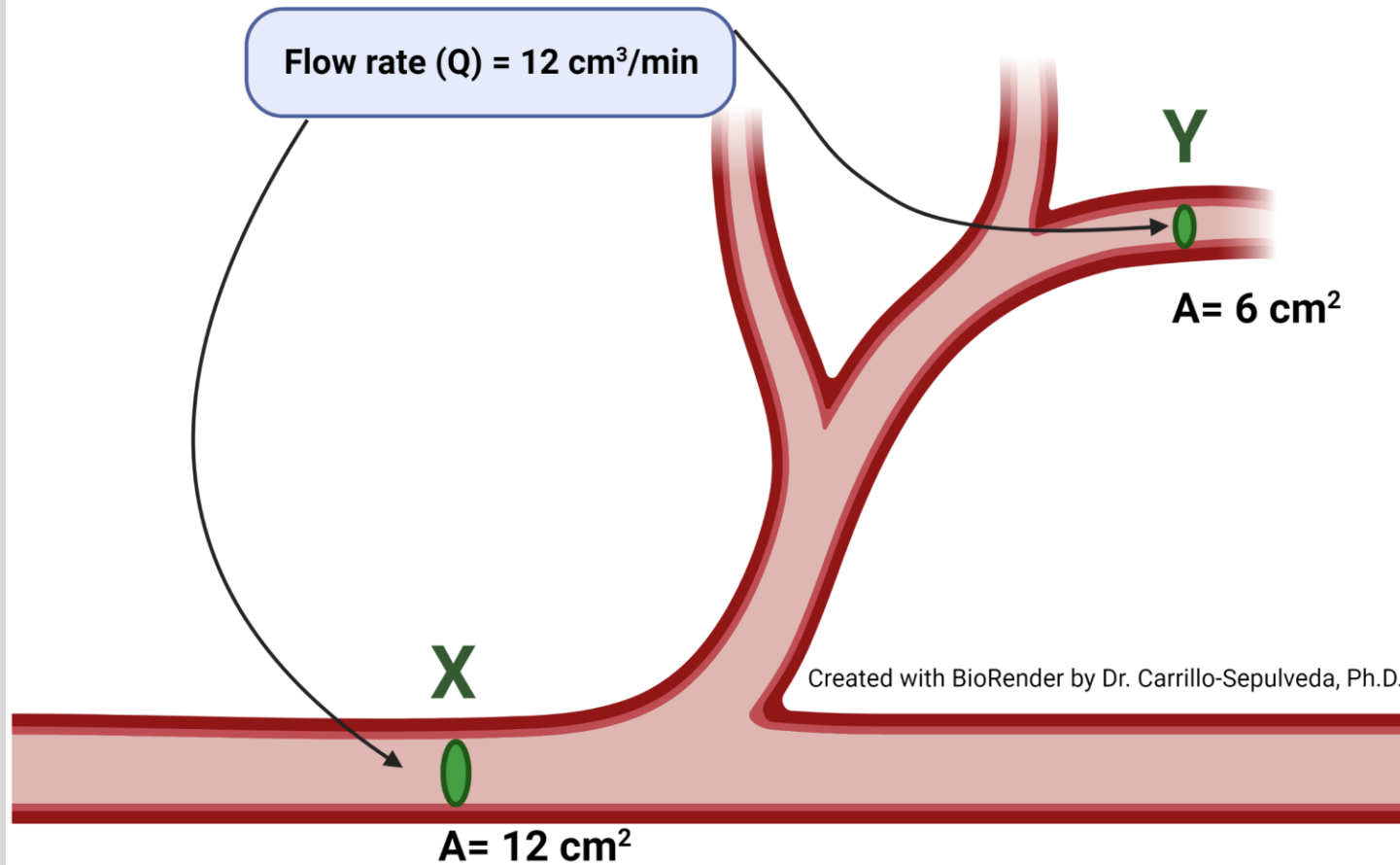
Compliance or **Capacitance** or **Distensibility** is the ability of a vessel to distend and increase volume with increasing transmural pressure. How much a vessel can increase the volume of blood, how much it can hold.



Velocity of Blood Flow

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The narrower the vessel,
the faster the velocity of
flow

$$\text{Velocity (v)} = \frac{\text{Flow rate (Q)}}{\text{Cross-section area (A)}}$$

$$\text{X} \quad v = \frac{12 \text{ cm}^3/\text{min}}{12 \text{ cm}^2} = 1$$

$$\text{Y} \quad v = \frac{12 \text{ cm}^3/\text{min}}{6 \text{ cm}^2} = 2$$

Because the velocity of blood flow is inversely proportional to vascular cross-sectional area, the changes in vessel diameter alter the velocity of blood flow through a vessel:

$$\text{Velocity (v)} = \frac{\text{flow (Q)}}{\text{area (A)}}$$

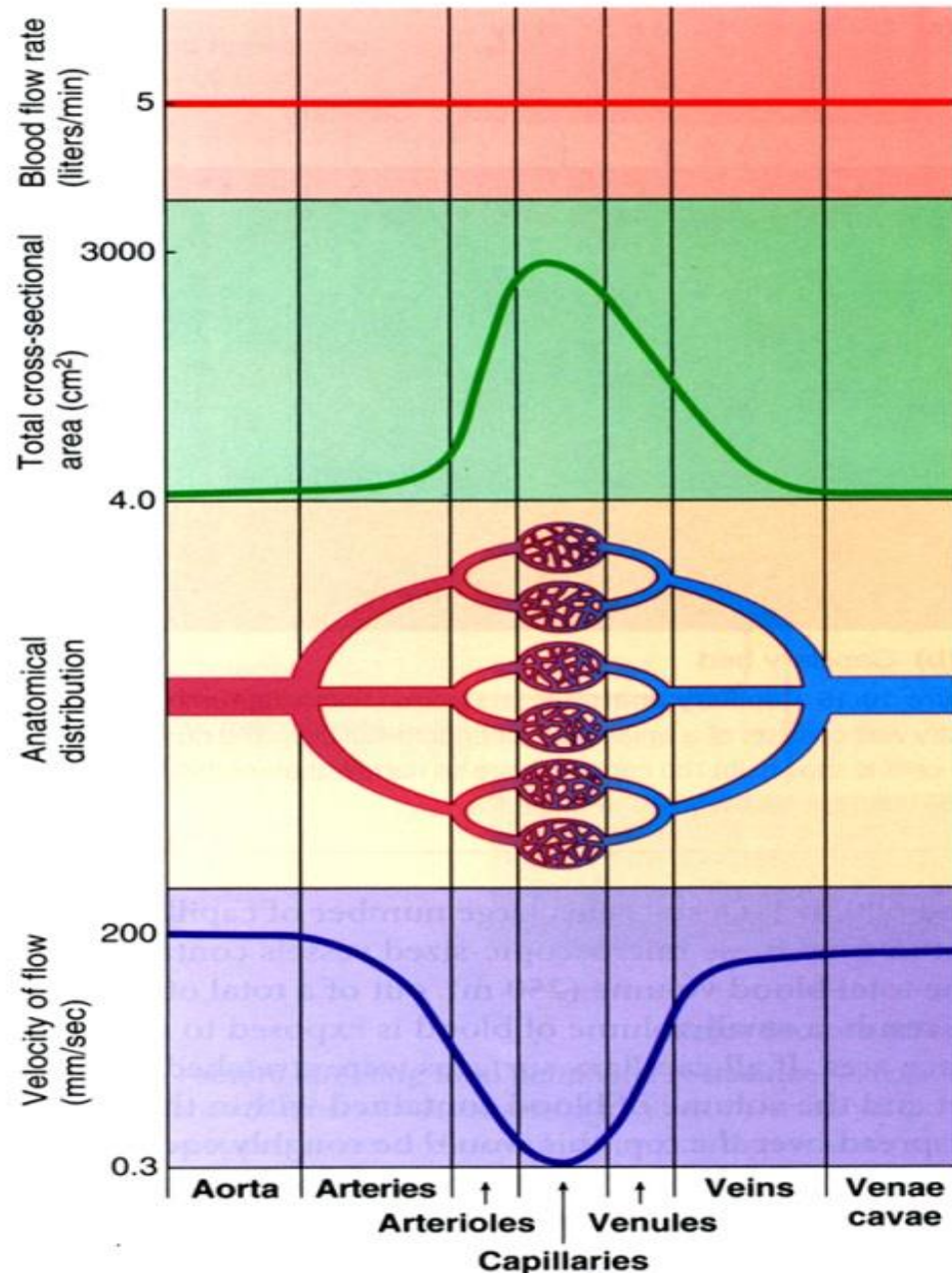
			
Area (A)	1cm ²	10cm ²	100cm ²
Flow (Q)	10 mL/sec	10 mL/sec	10 mL/sec
Velocity (v)	10 cm/sec	1 cm/sec	0.1 cm/sec

Vascular Cross-Sectional Area and Blood Flow

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$$\text{Velocity} = \frac{Q}{A} = \frac{Q}{\pi r^2}$$



♥ Flow rate remains **constant** because velocity changes with the changes in vessel radius.

♥ Capillaries have highest total cross-sectional area and lowest blood velocity

♥ *Velocity of blood flow is highest in the aorta and lowest in the capillaries because of a **dramatic increase in total cross-sectional area**. Velocity increases again as blood moves from the capillaries to the venules to veins, and finally, to the vena cava because total cross-sectional area is decreasing again.*

Practice Question

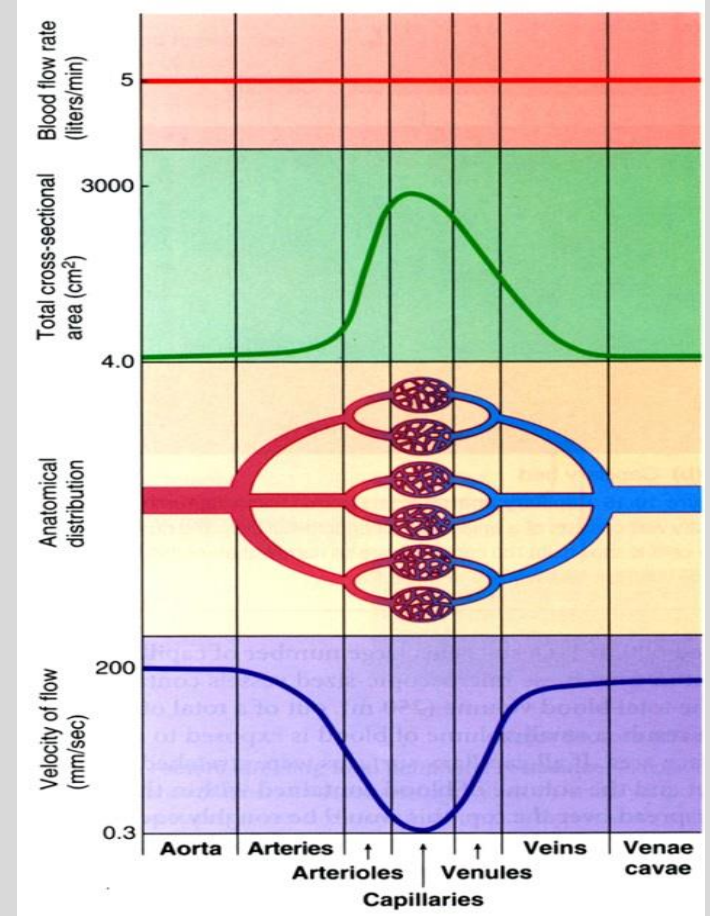
A man has a cardiac output of 5.5L/min. The diameter of his thoracic aorta is estimated to be 20mm, and the total cross-sectional area of his systemic capillaries is estimated to be 2500 cm². What is the velocity of blood flow in the aorta relative to the velocity of blood flow in the capillaries?

Capillaries: Velocity = $Q/A = 5.5\text{L/min} / 25000 = 2.2 \text{ cm/min}$

$$\text{Area} = \pi r^2$$

$$\text{Area} = 3.14 \times (10\text{mm})^2 = 3.14 \times (1\text{cm})^2 = 3.14 \text{ cm}^2$$

Aorta: Velocity = $Q/A = 5.5\text{L/min} / 3.14 = 1752\text{cm/min}$



Summary:

To keep **constant blood flow**, we need to take into consideration: cross section area of the vessels, and velocity

♥ Velocity ✓

Next:

- ♥ Pressure
- ♥ Resistance

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$$\text{Velocity} = \frac{Q}{A} = \frac{Q}{\pi r^2}$$

Blood Flow: Pressure and Resistance

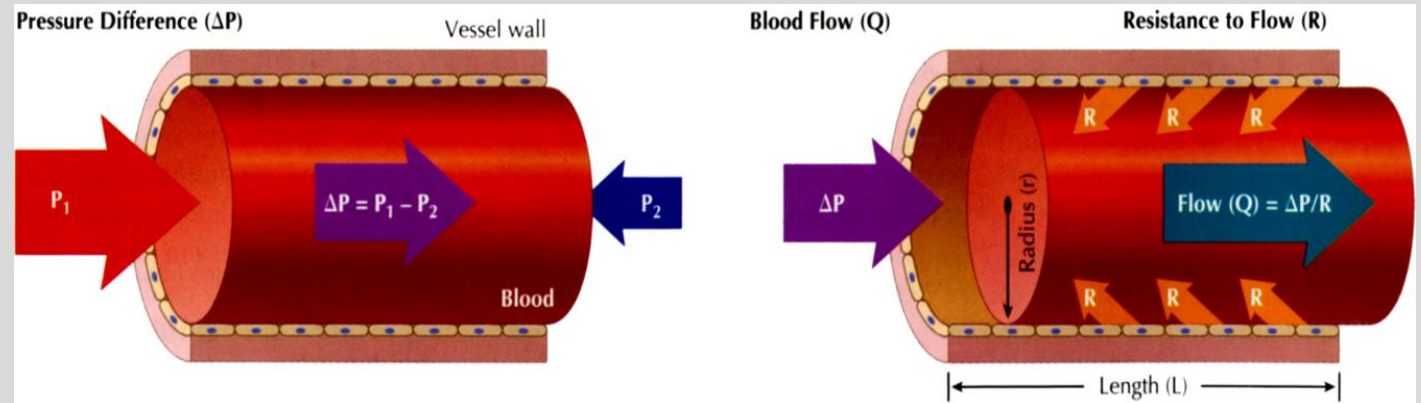
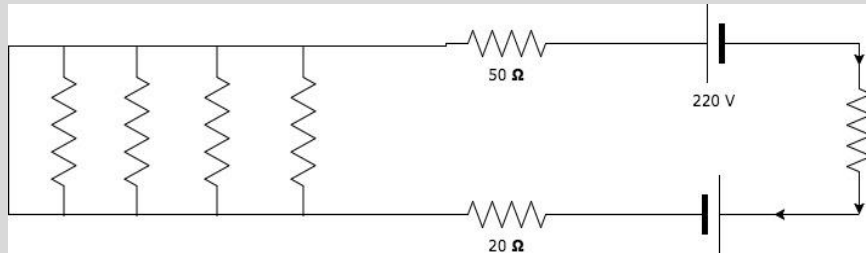
Ohm's Law, fundamental law of physics, states that current (I) equals the voltage difference (ΔV) divided by resistance (R). In relating Ohm's Law to fluid flow, the voltage difference is the **pressure difference** (ΔP ; sometimes called **driving pressure**, **perfusion pressure**, or **pressure gradient**), the resistance is the resistance to flow (R) offered by the blood vessel and its interactions with the flowing blood, and the current is the blood flow (Q).

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Electrical Circuit - Ohm's Law : $I = \frac{\Delta \text{Voltage}}{\text{Resistance}}$

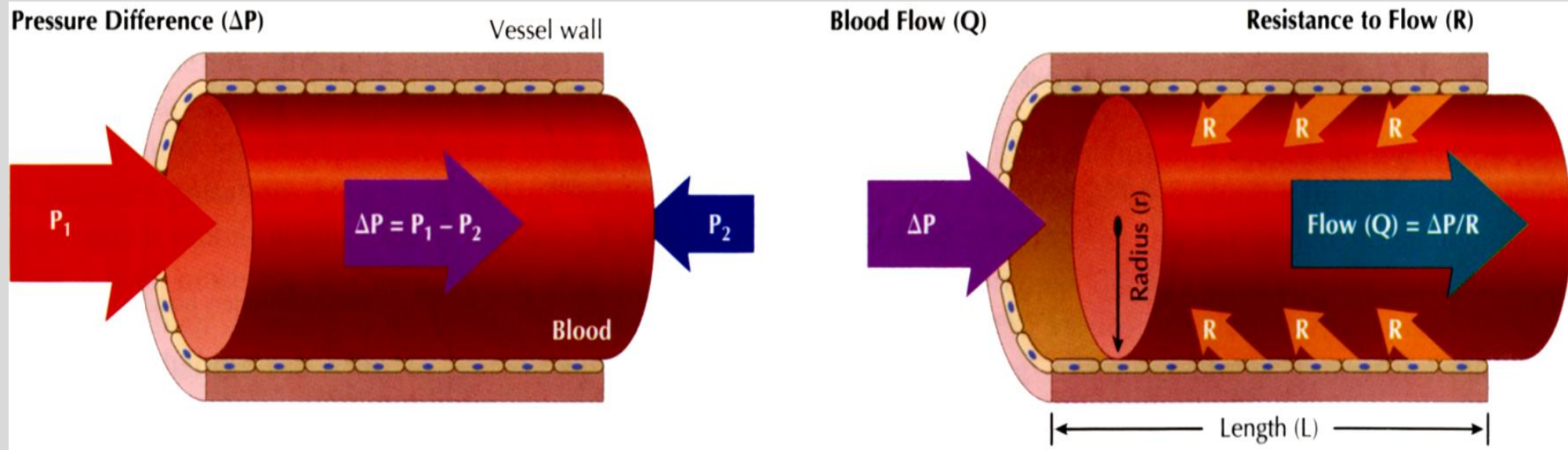
Blood flow (Q): $Q = \frac{P_1 - P_2}{R} = \frac{\Delta P}{R}$ (mL/min)



The major mechanism for changing blood flow in the cardiovascular system is by changing the resistance of blood vessels, particularly the arterioles

Blood Flow, Pressure and Resistance

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Pressure Difference (ΔP) is estimated as pressure difference of the blood between the two ends of the vessel segment: $P_1 - P_2 = \Delta P$ (mmHg). Flow is always from higher (P_1) to lower (P_2) pressure, from arteries to veins.

Vascular resistance (R) to blood flow occurs as a *result of friction between the flowing blood and the intravascular endothelium* along the inside of the vessel.

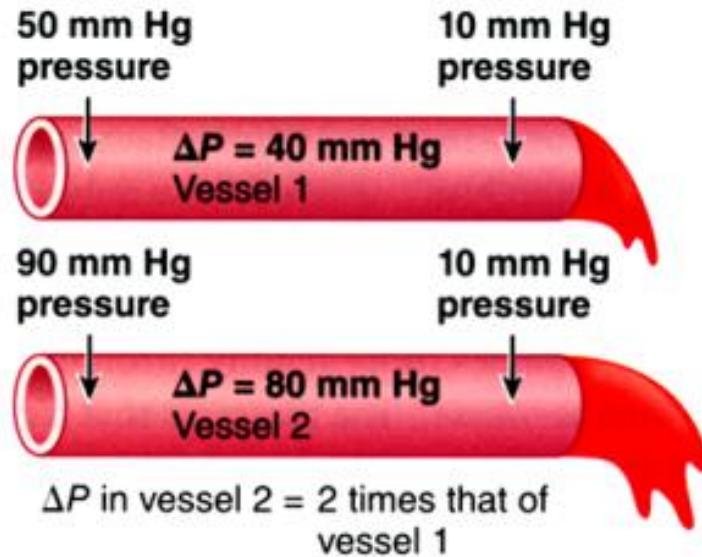
Blood Flow is directly proportional to the pressure difference but inversely proportional to the resistance:

$$Q = \frac{P_1 - P_2}{R} = \frac{\Delta P}{R} \text{ (mL/min)}$$

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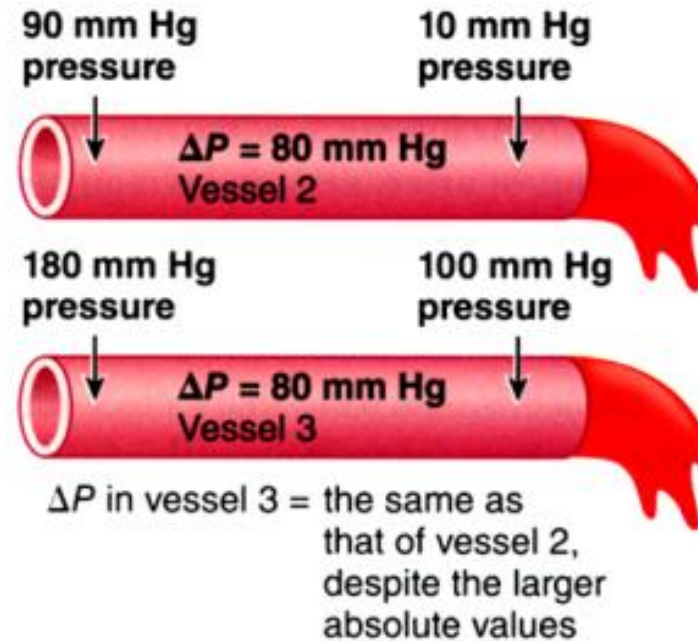
(a) Comparison of flow rate in vessels with a different ΔP



Flow in vessel 2 = 2 times that of vessel 1

Flow $\propto \Delta P$

(b) Comparison of flow rate in vessels with the same ΔP



Flow in vessel 3 = the same as that of vessel 2

Flow $\propto \Delta P$

Question

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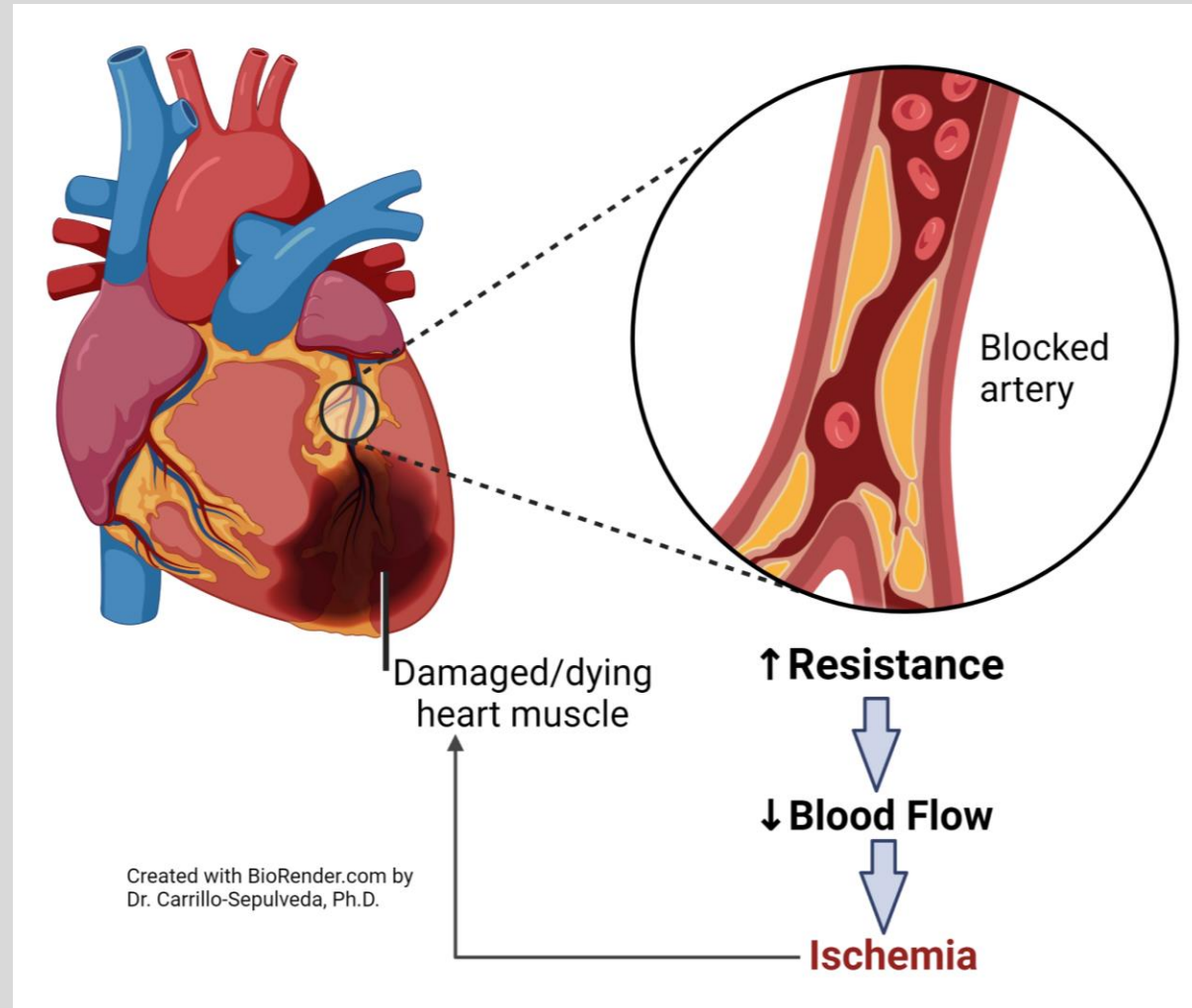


Ohm's Law

$$Q = \frac{P_1 - P_2}{R} = \frac{\Delta P}{R} \text{ (mL/min)}$$

The equation is flanked by two red arrows: a downward arrow on the left and an upward arrow on the right.

Ischemia



Practice Question

A 55-year-old-woman has renal blood flow of 500mL/min. Her renal arterial pressure is 100 mmHg and renal venous pressure is 10 mmHg. What is the vascular resistance of the left kidney in this woman?

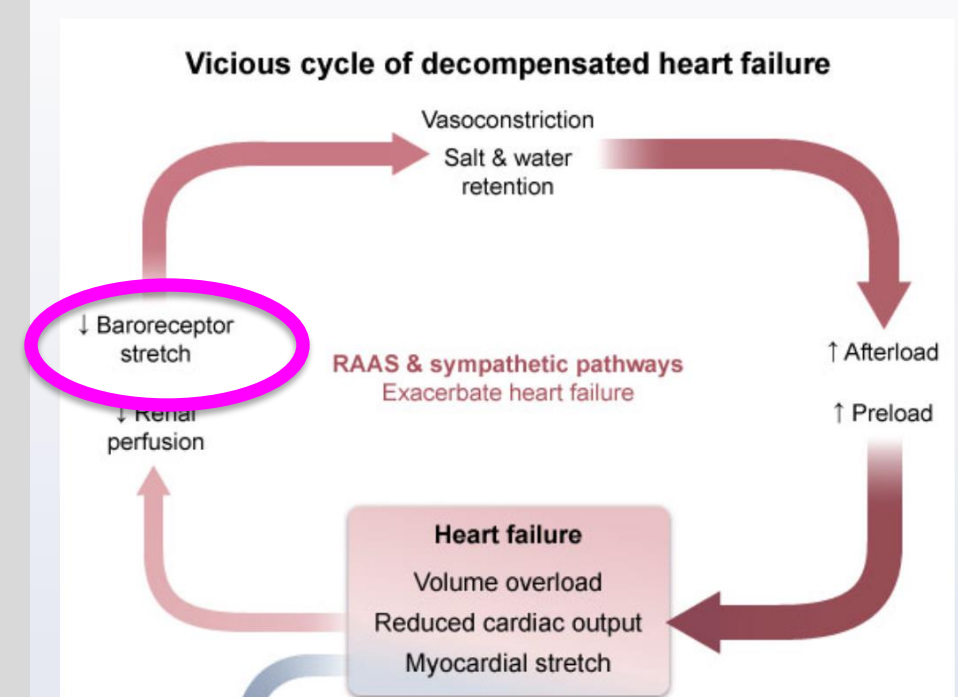
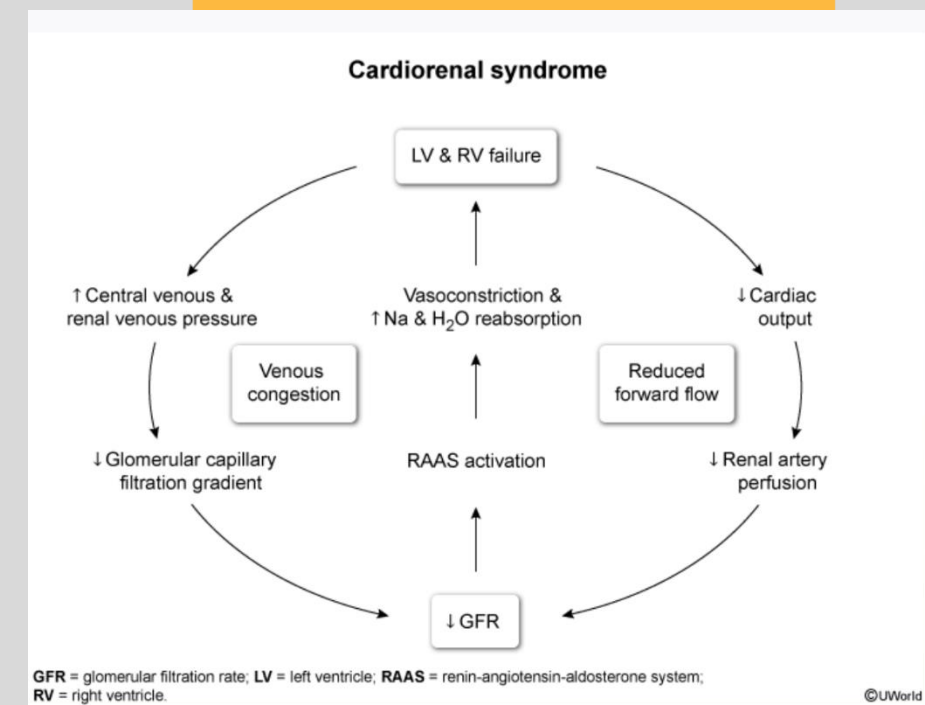
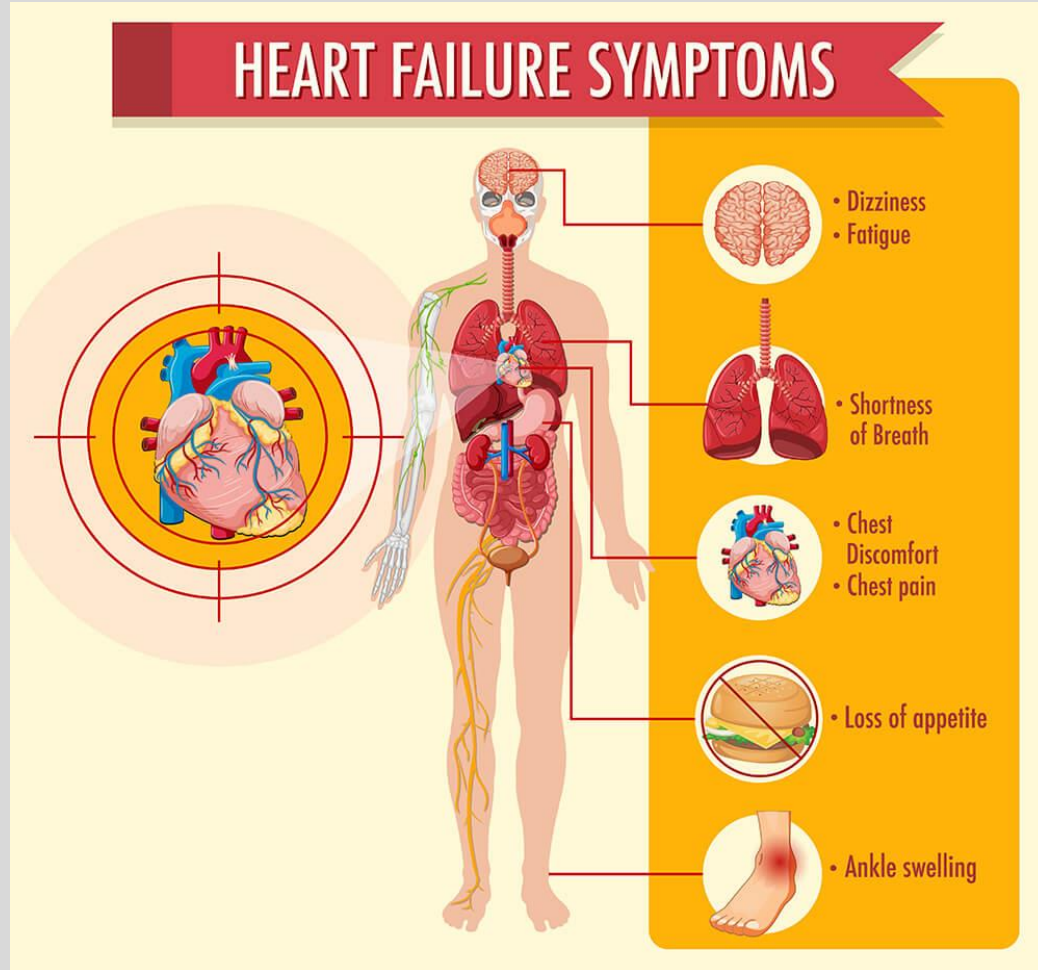
$$Q = \frac{\Delta P}{R}$$

$$500 = \frac{100 - 10}{R}$$

$$R = \frac{90}{500} = 0.18 \text{ mmHg/ml/min}$$

Practice Question

What most likely increase in a patient with decompensated heart failure?



Next:

- ♥ Velocity ✓
- ♥ Pressure ✓
- ♥ Resistance

The relationship between resistance, blood vessels diameter (radius) and blood viscosity is described by the Poiseuille Equation

Blood Flow and Resistance

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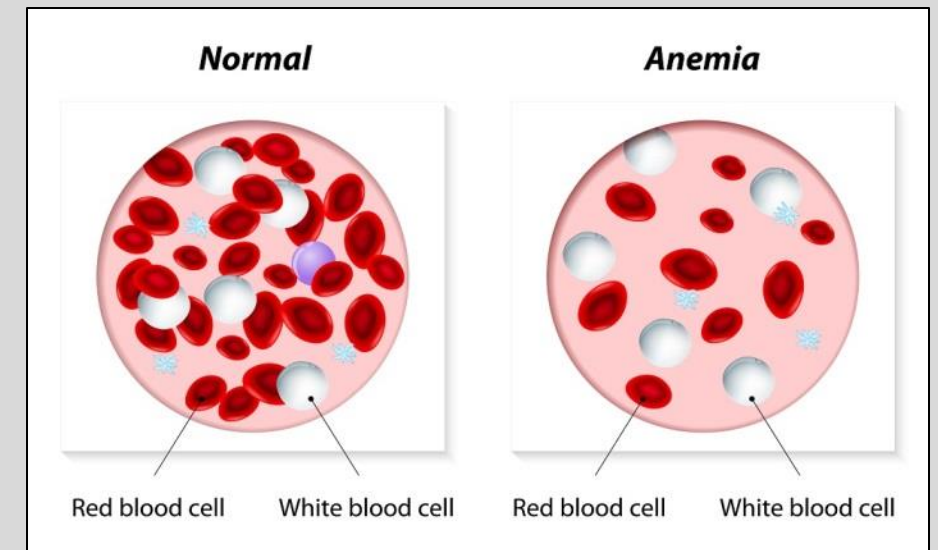
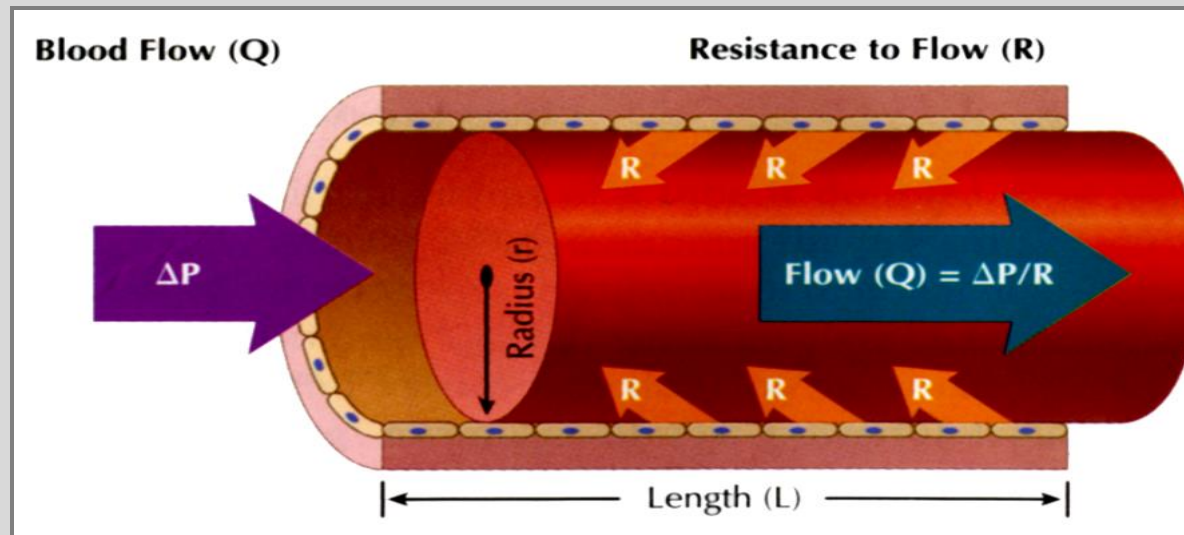
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$$\text{Flow (Q)} = \frac{\Delta P}{R}$$

$$R = \frac{8 L \eta}{\pi r^4}$$

$$\text{Flow (Q)} = \frac{\Delta P \pi r^4}{8 L \eta}$$

Poiseuille's Equation



Resistance, pressure, flow

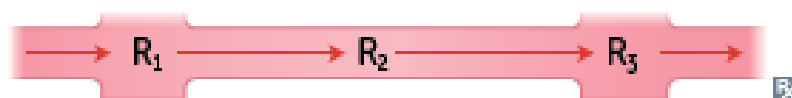
Volumetric flow rate (Q) = flow velocity (v) $\times$ cross-sectional area (A)

Resistance

$$= \frac{\text{driving pressure } (\Delta P)}{Q} = \frac{8\eta \text{ (viscosity)} \times \text{length}}{\pi r^4}$$

Total resistance of vessels in series:

$$R_T = R_1 + R_2 + R_3 \dots$$



Total resistance of vessels in parallel:

$$\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} \dots$$



$$Q \propto r^4$$

$$R \propto 1/r^4$$

Capillaries have highest total cross-sectional area and lowest flow velocity.

Pressure gradient drives flow from high pressure to low pressure.

Arterioles account for most of TPR. Veins provide most of blood storage capacity.

Viscosity depends mostly on hematocrit.

Viscosity $\uparrow$ in hyperproteinemic states (eg, multiple myeloma), polycythemia.

Viscosity $\downarrow$ in anemia.

Practice Question

A 65-year-old man with peptic ulcer disease comes to the emergency department due to nausea, melena and feels faint. His blood pressure is 90/40 mmHg, pulse is 130/min, and respirations are 28/min. Physical examination reveals a lethargic man with cold extremities. The patient receives blood transfusion; however, the transfusion rate is very slow, and the patient remains hypotensive. Which of the following will lead to the greatest improvement in flow rate?

- a. Doubling the diameter of the intravenous catheter
- b. Doubling the length of the intravenous catheter
- c. Doubling the pressure of the infusion pump
- d. Placing a second intravenous catheter with identical diameter and length

Poiseuille's Equation

$$\text{Flow (Q)} = \frac{\Delta P \pi r^4}{8L\eta}$$

Hypovolemic Hemorrhagic Shock

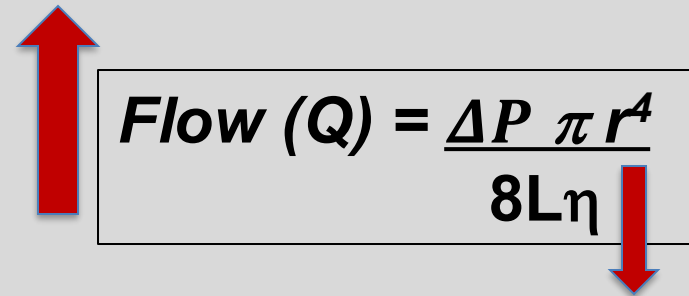
Practice Question

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A 22-year-old medical student presents with a 3-month history of fatigue and complain of feeling sleepy most of the time. She is found to have a mid-systolic murmur and higher than normal cardiac output. The differential diagnosis based on these findings includes which of the following?

- a.) Cardiac tamponade
- b.) Mitral regurgitation
- c.) Third-degree heart block
- d.) Anemia



The diagram shows the equation for flow (Q) in a tube, enclosed in a black rectangular box. A large red arrow points upwards from the bottom left of the box, and another large red arrow points downwards from the bottom right of the box. The equation inside the box is:

$$\text{Flow (Q)} = \frac{\Delta P \pi r^4}{8L\eta}$$

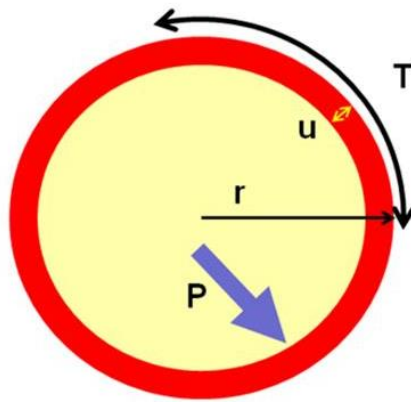
La Place Law- Wall Tension in a Blood Vessel

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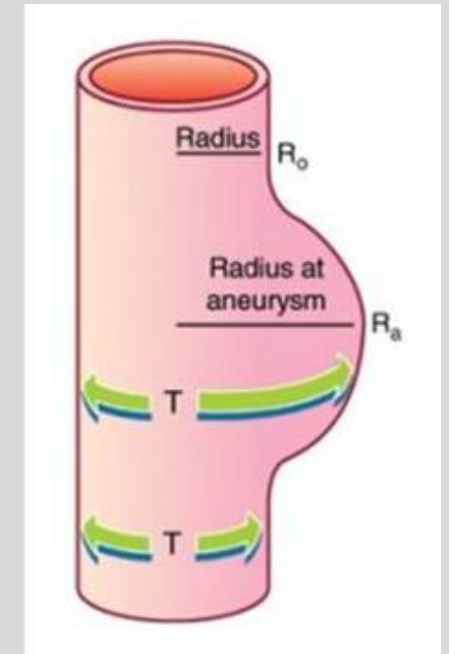
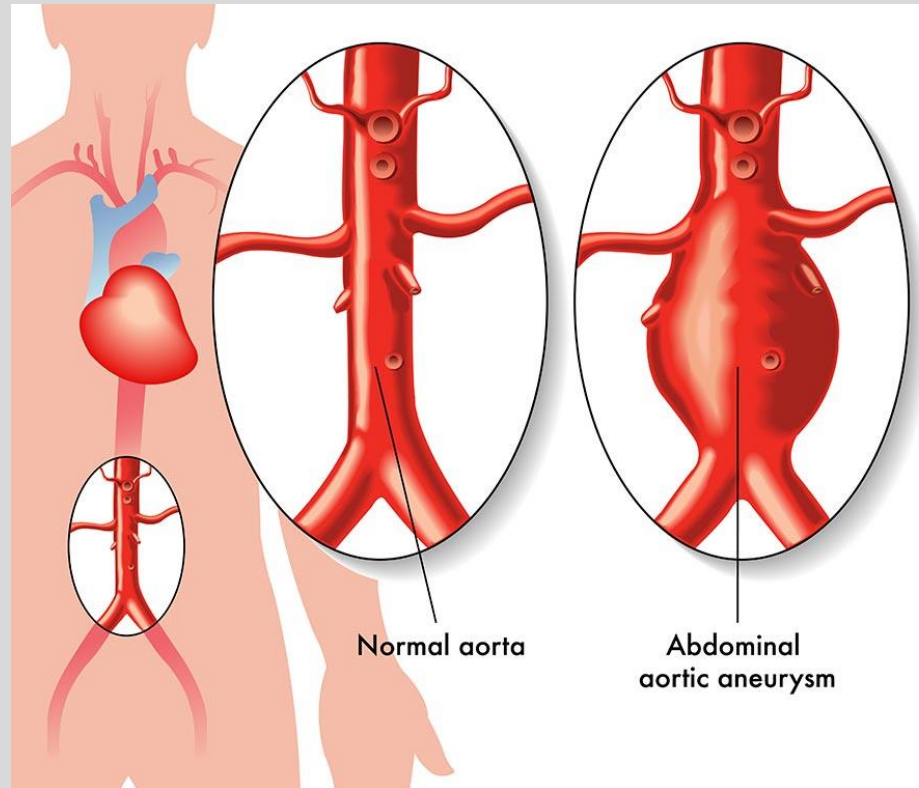
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Why does the wall tension increase with the radius

LaPlace's Law



$$\text{Wall Tension (T)} = \frac{\text{Transmurial Pressure (P)} \times \text{Radius (r)}}{2 \times \text{Wall Thickness (u)}}$$

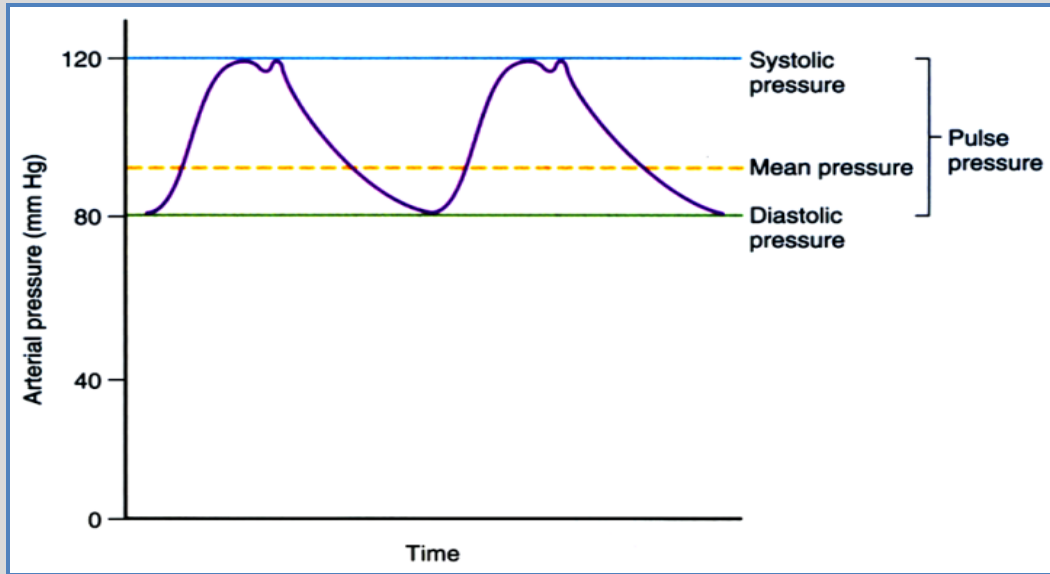


- ♥ Arterial Compliance
- ♥ Arterial Stiffness
- ♥ Arterial Thickness

Systemic Arterial Pressures during the Cardiac Cycle

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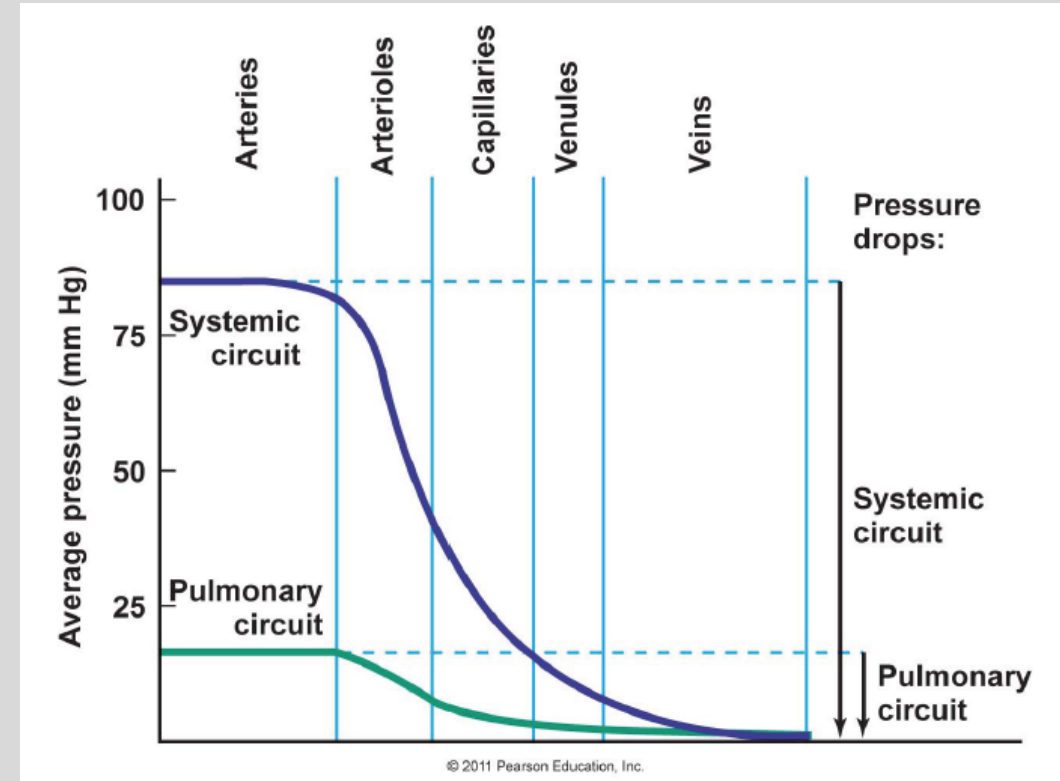
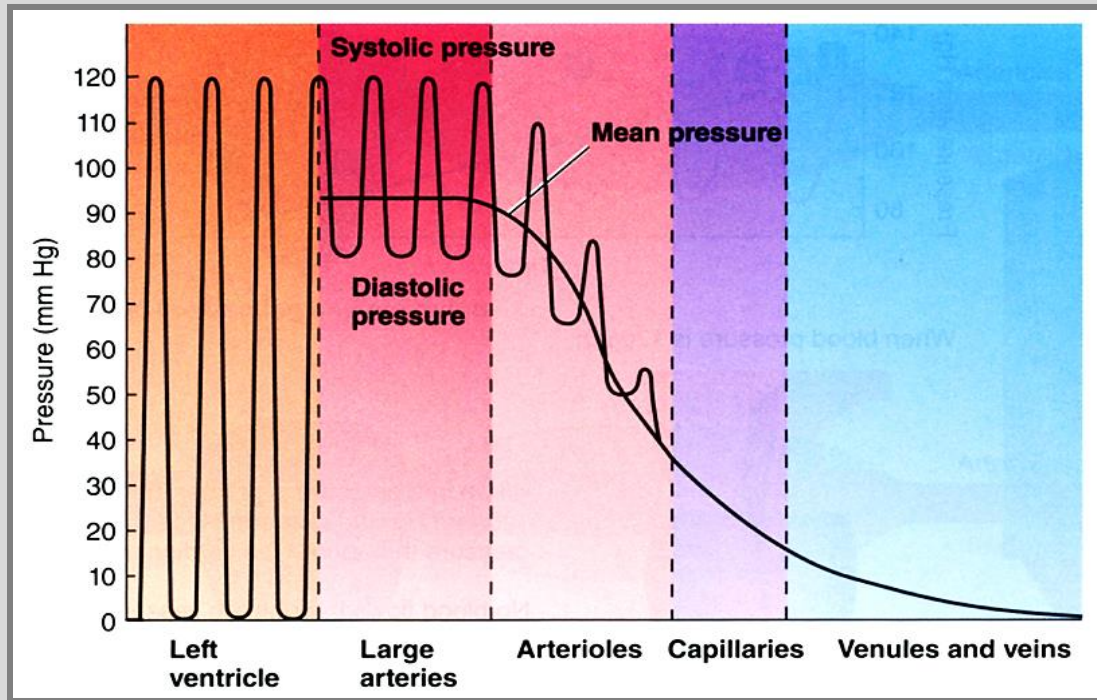


$$\text{Mean Arterial Pressure} = \text{Diastolic Pressure} + \left(\frac{1}{3} \times \text{Pulse Pressure} \right)$$

$$\text{Pulse Pressure} = \text{Systolic Pressure} - \text{Diastolic Pressure}$$

- ♥ **Systolic Pressure** is the *highest arterial pressure* measured during a cardiac cycle.
 - ♥ **Diastolic Pressure** is the *lowest arterial pressure* measured during a cardiac cycle.
 - ♥ **Mean Arterial Pressure** is the running average of arterial pressure in the body
 - ♥ **Pulse Pressure** is the *difference between systolic pressure and diastolic pressure*.
- The *magnitude of the pulse pressure reflects the volume of blood ejected on a single beat*.

Pressure profile in the Vasculature



Systemic Circuit – Avg. Systolic Pressure (120-125 mmHg)

Avg. Diastolic Pressure (60-80 mmHg)

Pulmonary Circuit - Avg. Systolic Pressure (20-24 mmHg)

Avg. Diastolic Pressure (4-8 mmHg)

Definitions

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For the entire systemic circulation:

Pulse Pressure: Systolic Pressure – Diastolic Pressure

Blood Pressure (BP): Cardiac Output x Total Peripheral Resistance

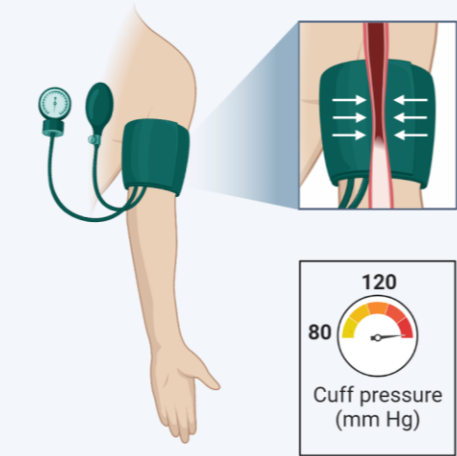
Total Peripheral Resistance (TPR) = Systemic Vascular Resistance (SVR)

Mean Arterial Pressure (MAP): $\frac{2 \times \text{Diastolic Pressure} + \text{Systolic Pressure}}{3}$

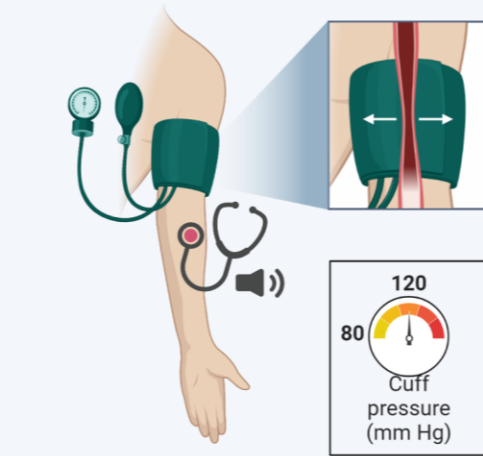
Cardiac Output: $\frac{\text{Aorta Pressure} - \text{Vena Cava Pressure}}{\text{Total Peripheral Resistance}}$

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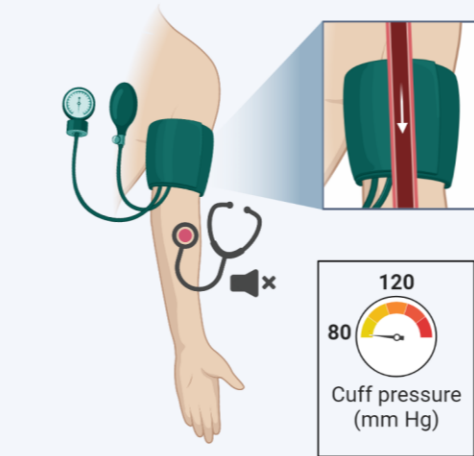
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- ① The blood pressure cuff is wrapped around the arm and inflated until blood flow is stopped.



- ② The examiner listens for arterial sounds as the cuff is deflated. Pressure at which sounds are first heard is recorded as the **systolic** pressure.



- ③ The pressure at which sounds are no longer heard is recorded as the **diastolic** pressure.



Systolic blood pressure: 120 mm Hg

Diastolic blood pressure: 80 mm Hg

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Dr. Carrillo-Sepulveda, Ph.D.

$$\text{Mean Arterial Pressure} = \text{Diastolic Pressure} + \left(\frac{1}{3} \times \text{Pulse Pressure} \right)$$

$$\text{Pulse Pressure} = \text{Systolic Pressure} - \text{Diastolic Pressure}$$

$$\text{Mean Arterial Pressure (MAP):} \\ \frac{2 \times \text{Diastolic Pressure} + \text{Systolic Pressure}}{3}$$

Pulse Pressure : $120 - 80 = 40$ mmHg

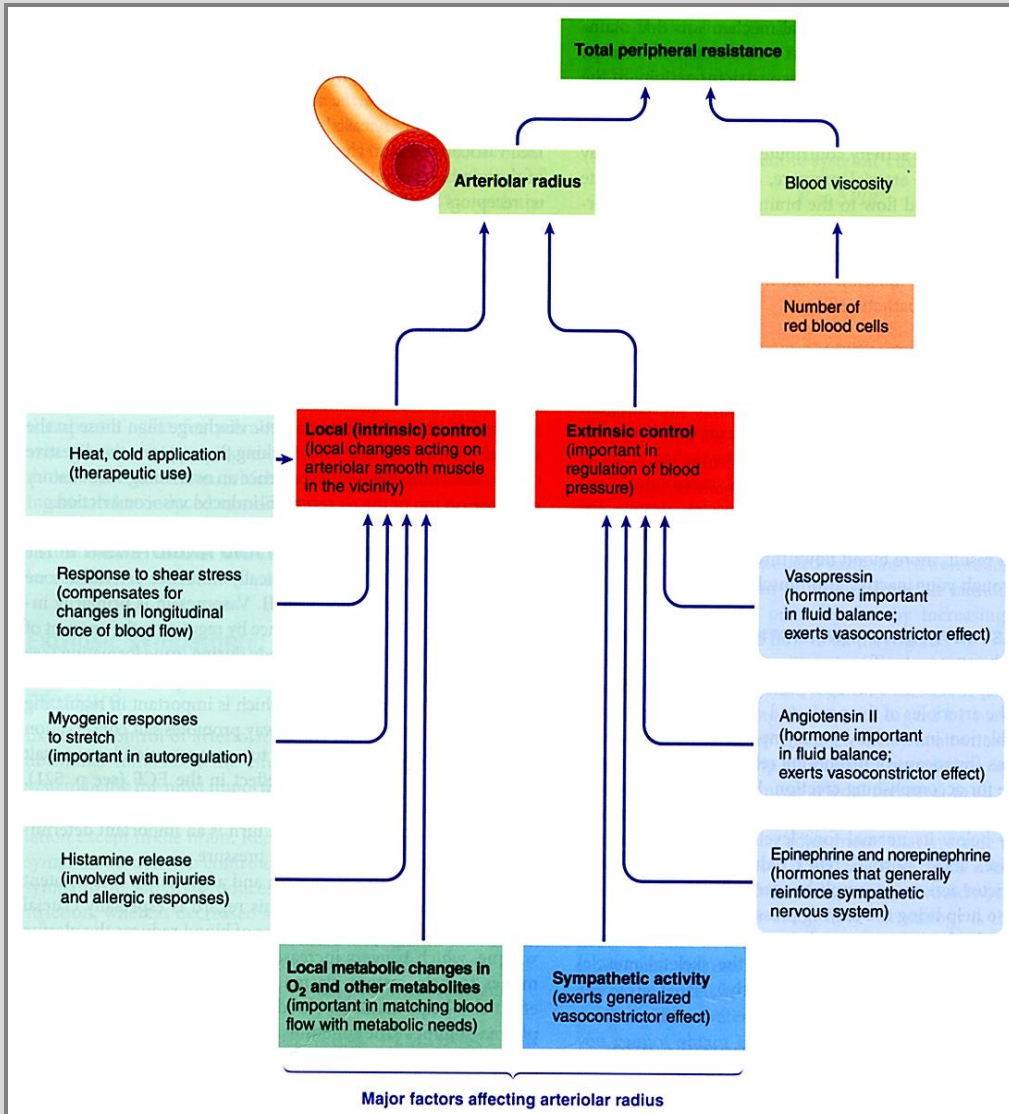
MAP: $80 + \frac{1}{3} \times 40 = 93$ mmHg

MAP: $2 \times 80 + 120 / 3 = 93$ mmHg

Normal Pulse Pressure = 30-40 mmHg

Total Peripheral Resistance

Resistance of the entire systemic vasculature is called the total peripheral resistance (TPR).



$$R = \frac{\Delta P}{Q}$$

$$TPR = \frac{\Delta P}{CO}$$

$$TPR = \frac{MAP - CVP}{CO}$$

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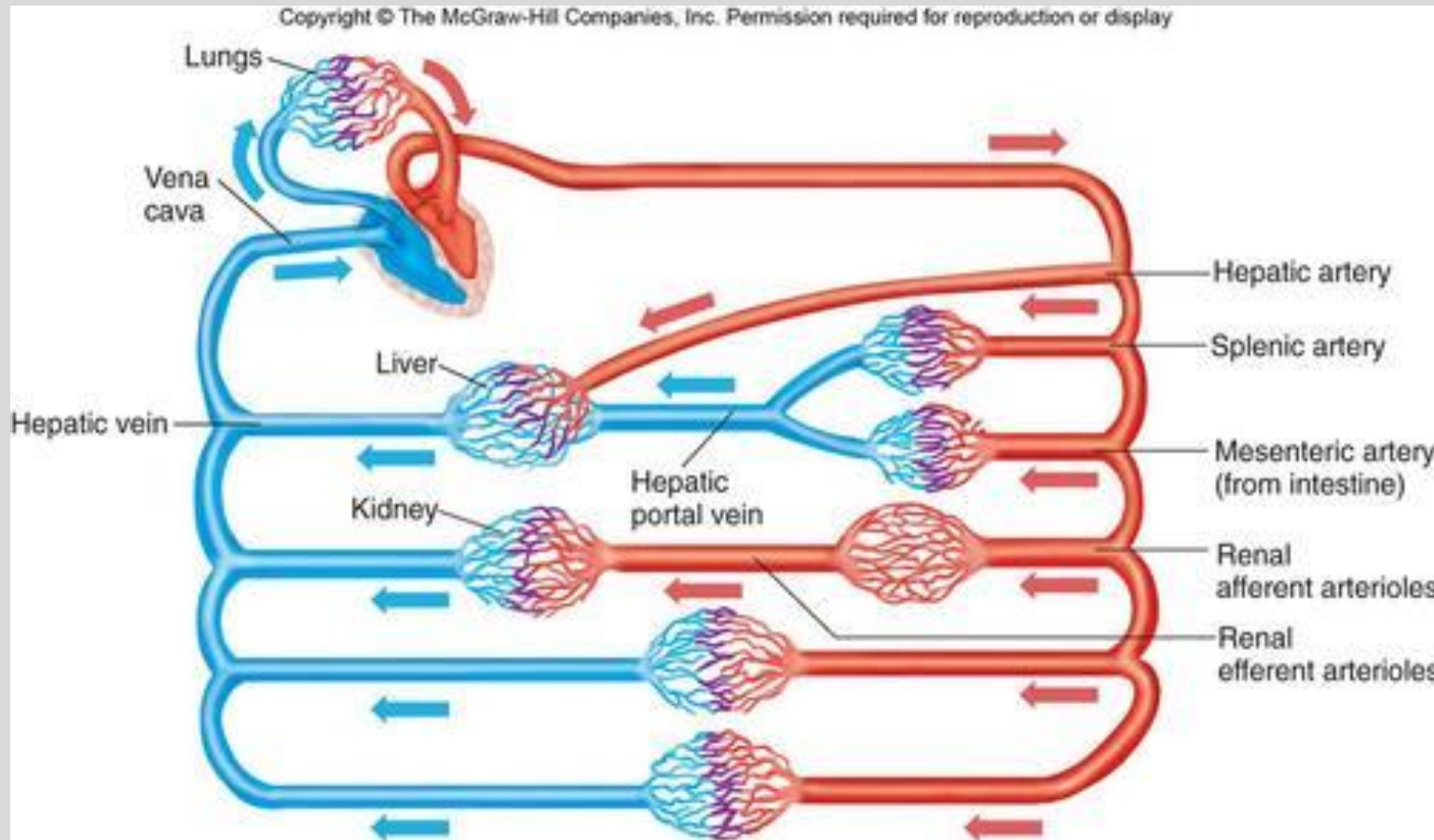
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- The primary determinant of TPR is the *adjustable arteriolar radius*.
- Two major categories of factors influence arteriolar radius:
 - 1) *Local* (intrinsic) control
 - 2) *Extrinsic* control

Total (Circulatory) Peripheral Resistance

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$$R = \frac{\Delta P}{Q}$$

$$TPR = \frac{\Delta P}{CO}$$

$$TPR = \frac{MAP - CVP}{CO}$$

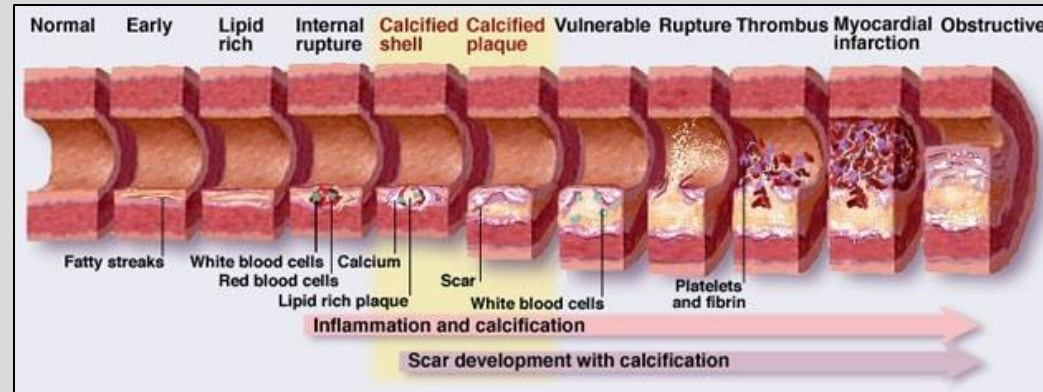
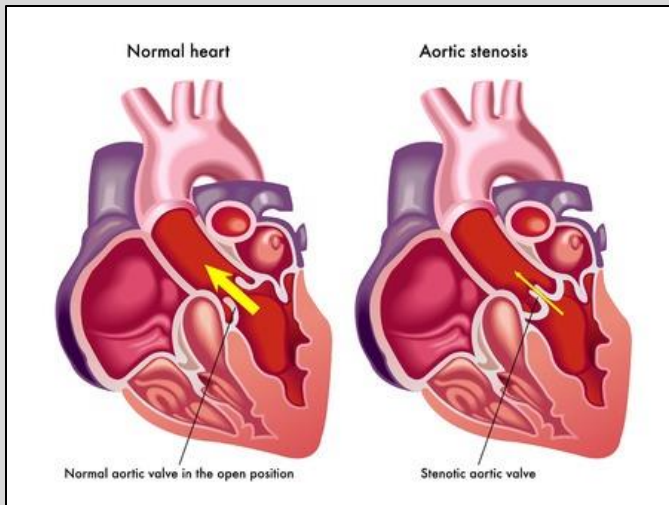
Total Peripheral Resistance: Mean Arterial Pressure – Central Venous Pressure
Cardiac Output

Pulse Pressure during Pathological Conditions

Pulse Pressure: Systolic Pressure – Diastolic Pressure

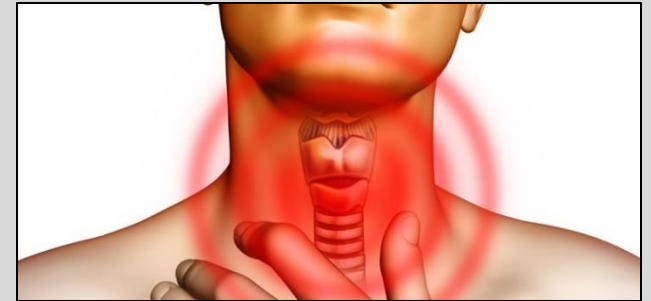
Factors implicated in the magnitude of the pulse pressure:

1. Stroke volume
2. Speed of ejection of the stroke volume (Ventricular Contraction Rate)
3. Arterial compliance (Stretch-distensibility)



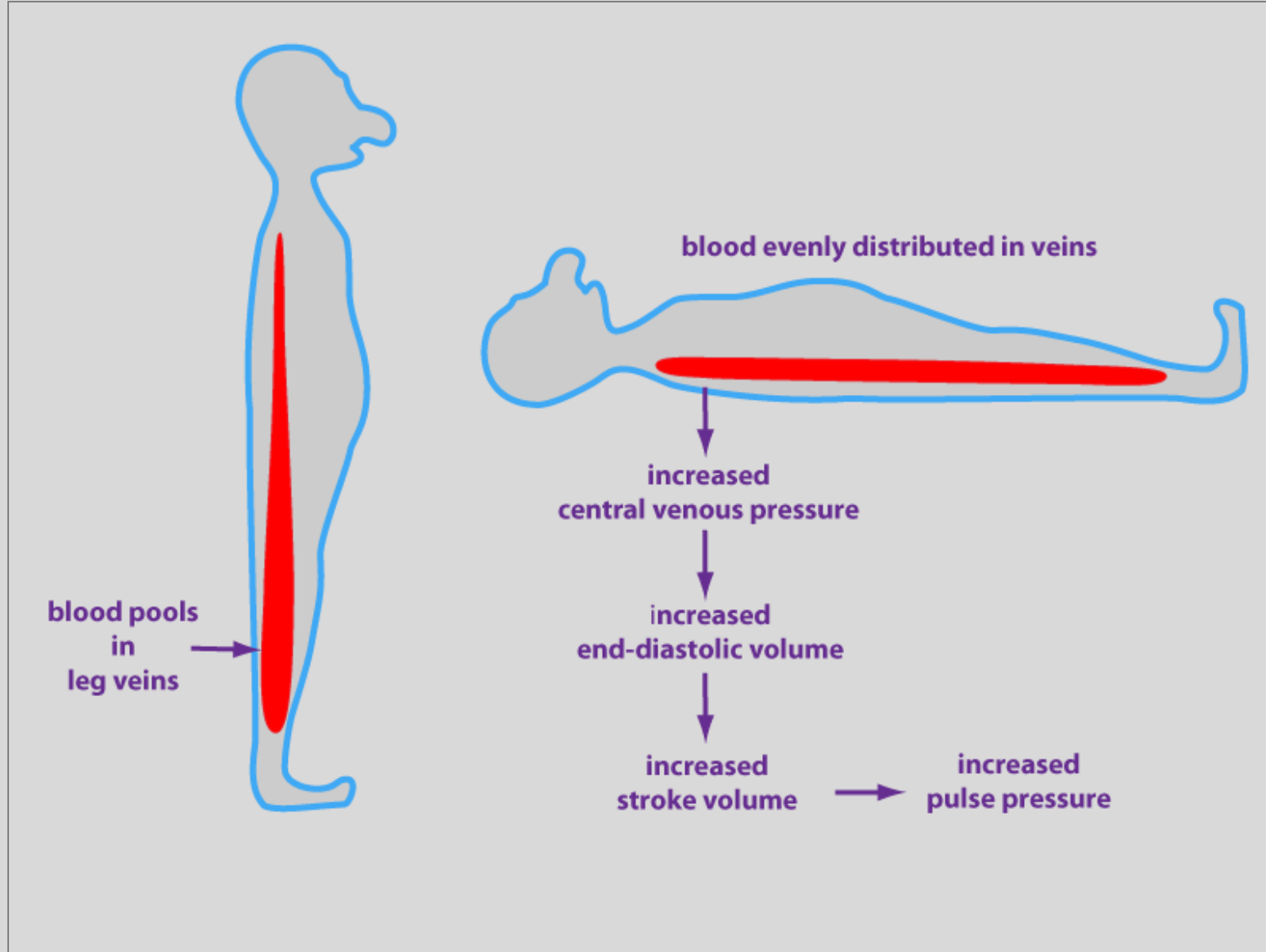
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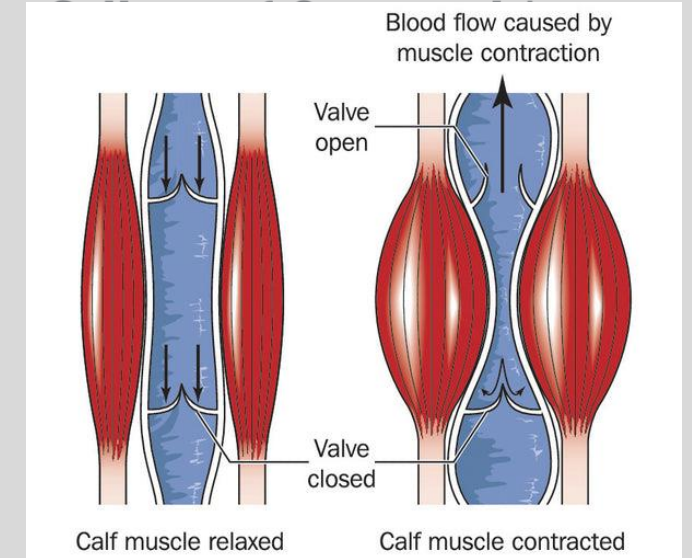


Clinical Relevance - Pressure

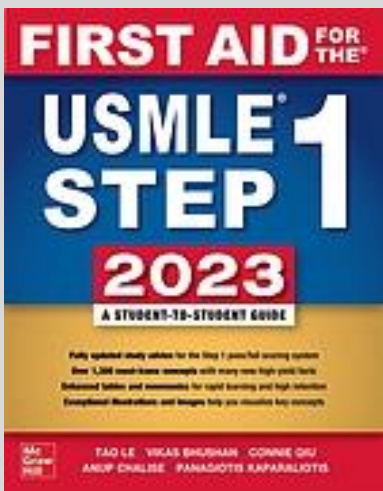
Standing to Supine Position



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<https://leglux.com/blogs/news/what-are-varicose-veins-what-causes-and-prevents-them>



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Pulse pressure

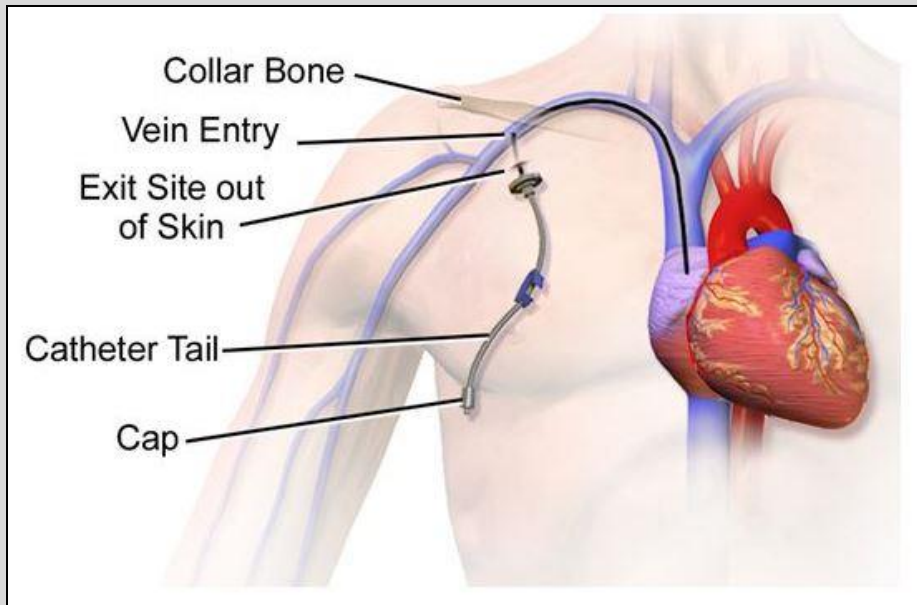
PP = systolic blood pressure (SBP) – diastolic blood pressure (DBP)

PP directly proportional to SV and inversely proportional to arterial compliance.
↑ PP in hyperthyroidism, aortic regurgitation, aortic stiffening (isolated systolic hypertension in elderly), obstructive sleep apnea (↑ sympathetic tone), anemia, exercise (transient).
↓ PP in aortic stenosis, cardiogenic shock, cardiac tamponade, advanced HF.

Central Venous Pressure (CVP)

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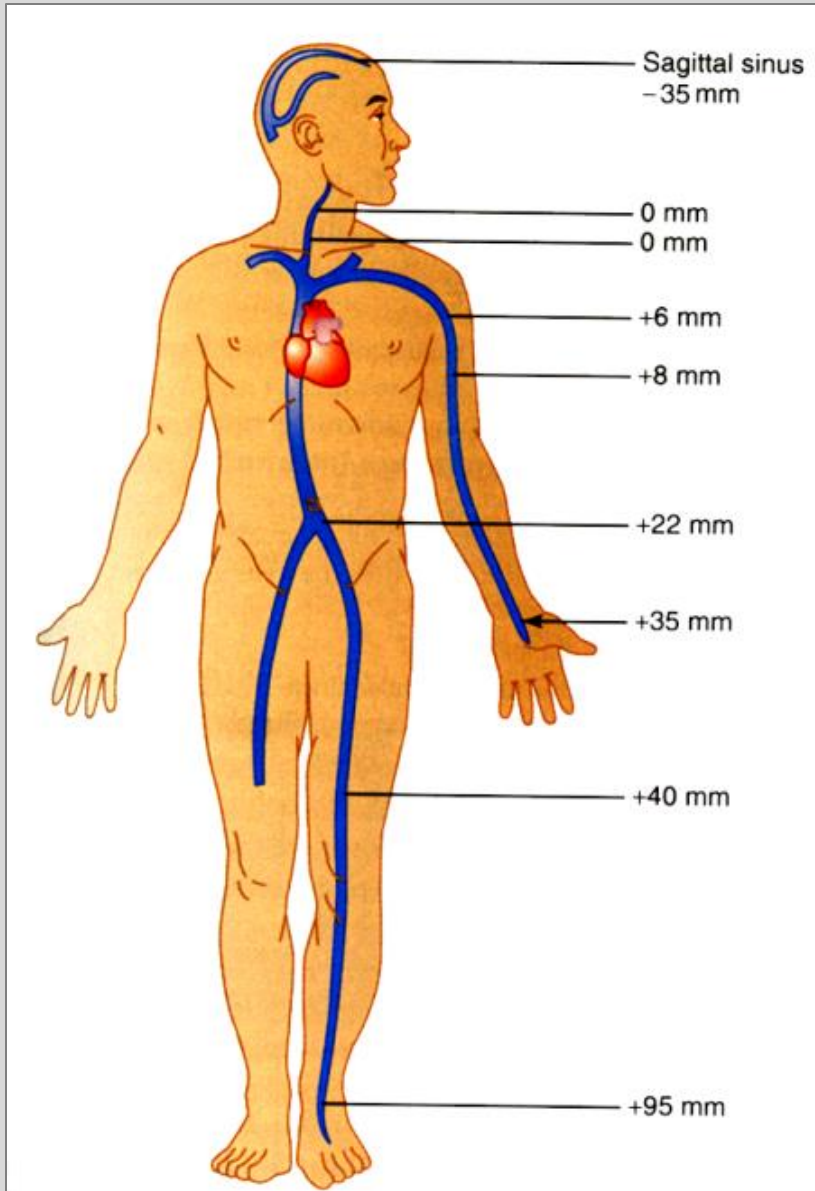
Blood pressure in thoracic vena cava, near right atrium

It is regulated by a balance between:

- the ability of the heart to pump blood out of the right atrium and ventricle into the lungs;
- the tendency for the blood to flow from the peripheral veins into the right atrium (venous return).

Normal Central Venous Pressure (CVP): 0-7 mmHg

- Weight of the blood in vessels is affected by gravitational force that generates hydrostatic pressure within the vascular system.



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- In a *standing adult*, *central venous pressure in the right atrium is ~0 mm Hg*, while *hydrostatic pressure in the veins of the feet can reach ~95 mm Hg* (depends on a person's height) *because of the weight of the blood in the veins* between the heart and the feet ($130 \text{ cm} / 13.6 \text{ mm} \approx 95 \text{ mm Hg}$).
- Gravitational venous *pressures at other levels* of the body, except neck and head, are *between ~0 and ~95 mm Hg*.
- If* there were *no one-way valves in veins*, the *hydrostatic pressure in a standing adult* would cause the venous pressure in the feet always to be *about 95 mm Hg*. However, under *ordinary circumstances*, the *venous pressure in the feet* remains *less than 30 mm Hg* because of the *leg skeletal muscle pump*.

Practice Question

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A 27-year-old medical student is sitting in silent study reviewing cardiovascular physiology. She takes her pulse and counts her heart rate to be 75 beats/min. Her friend preparing for ICC measures her blood pressure and tells her she has a systolic pressure of 110 mmHg and a diastolic pressure of 70 mmHg. She knows her average stroke volume is 65mL/beat. Based on this information:

What is her Cardiac Output?

What is her Mean Arterial Pressure?

What is her Pulse Pressure?

Cardiac Output - Calculations

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Physical Calculation

$$CO = SV \times HR$$

- Mechanical Calculation
- Pump/Engine
- Requires Testing

Fick's Principle

$$CO = \frac{[O_2 \text{ Consumption}]}{[O_2 \text{ Arterial}] - [O_2 \text{ Venous}]}$$

- Physiologic Approximation
- Cardio-Pulmonary-Metabolic Assessment
- Requires Labs

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Practice Question

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A 70-year-old man presents with the following parameters:

Heart Rate: 50 beats/min

Pulmonary vein O₂ content: 21mL O₂/100mL

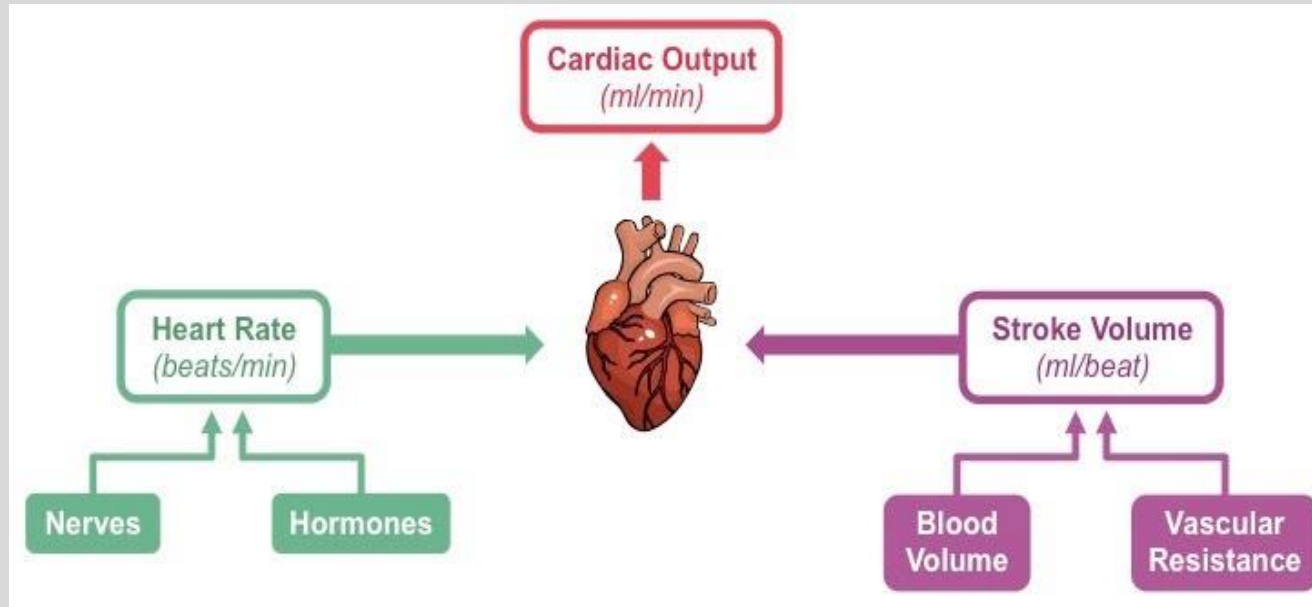
Pulmonary artery O₂ content: 6 mL O₂/100mL

Oxygen consumption (VO₂) : 300mL/min

Stroke volume: 40 mL

What is the patient's cardiac output?

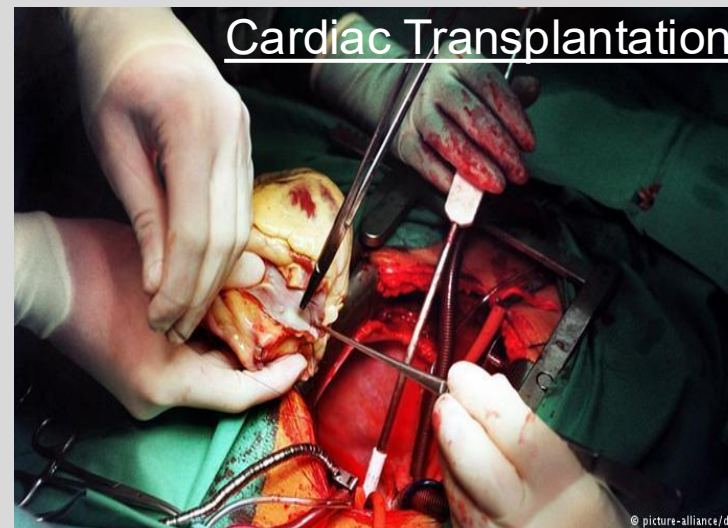
Alterations in Cardiac Output - Pump



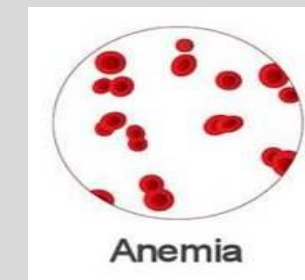
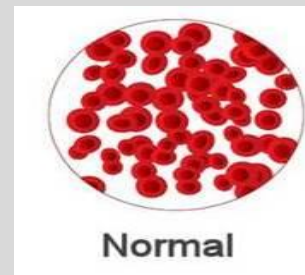
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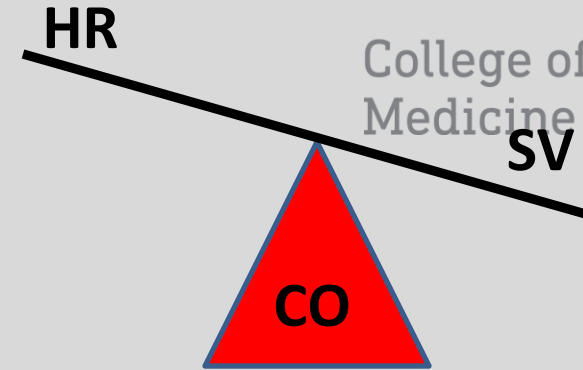
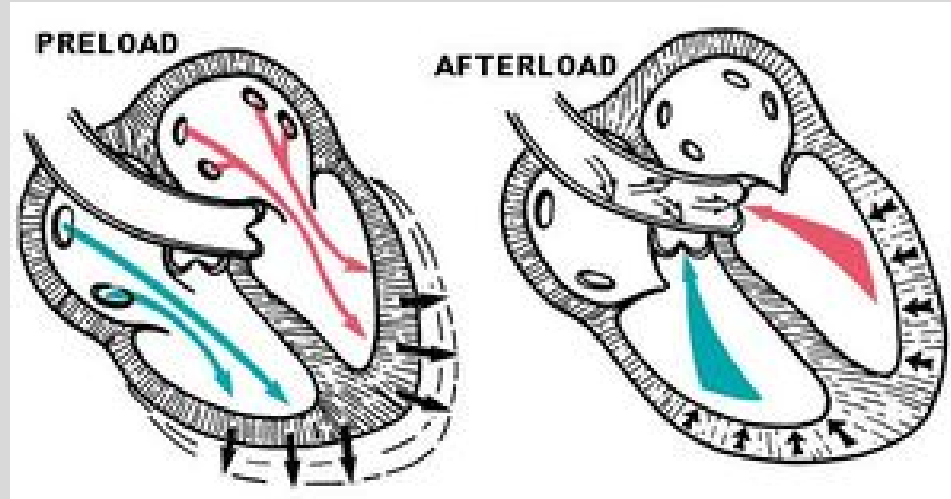
Hemodynamic function of an arrested heart



$CO = HR \times SV$ - Stroke Volume

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Stroke Volume (SV) is the volume of blood ejected by the ventricle in one contraction. (ml/beat)- ***Most commonly LV***

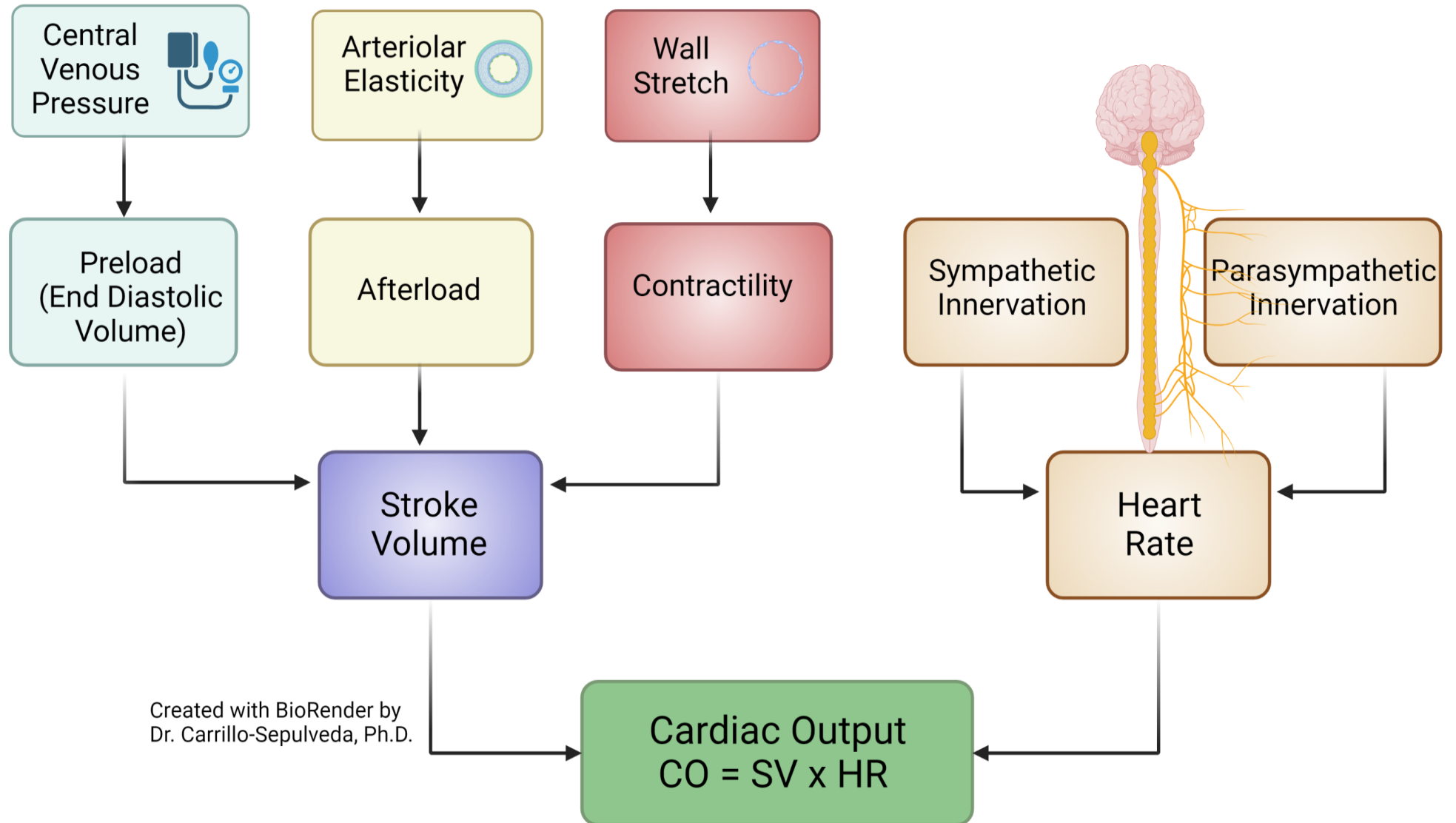
Preload is the degree of stretch on the myocardium at the **end of diastole (mL)**

Afterload is the **resistance against which blood is ejected (mmHg)**

↓ Stroke Volume:

- ♥ a decrease in preload
- ♥ an increase in afterload
- ♥ a decrease in cardiac contractility

Relationship of Cardiac Output, Stroke Volume and Heart Rate



Summary Question:

A 64 year old female presents to the office following a hospital visit for diabetic ketoacidosis. Her medical reports show that at the time of admission the patient had the following vitals signs: Heart rate 55 beats/min a respiratory rate of 27 breaths/min systolic blood pressure was 156 mmHg and diastolic pressure was 85 mmHg. Echocardiogram revealed a right atrial pressure of 1 mmHg. A venous Doppler revealed poor upper and lower extremity perfusion. Arterial blood gas results showed an arterial O_2 concentration of 24mL O_2 /100mL. Venous blood gas results showed a venous O_2 concentration of 18 mL O_2 /100mL. Metabolic oxygen consumption rate was determined to be 150mL/min. Based on these reports what was the patients systemic vascular resistance at the time of admission?

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